

RankUp NEET — Ultra Premium Sheet

Physics • Laws of Motion + Motion (1D & 2D) • Target: AIR < 100 • 20 Questions

Q1. A 5 kg block lies on a rough horizontal surface ($\mu = 0.4$). A horizontal force is increased gradually till the block just starts moving (take $g = 10 \text{ m/s}^2$). If the force is suddenly doubled from this value, the acceleration becomes:

- A. 0 m/s^2
- B. 2 m/s^2
- C. 4 m/s^2
- D. 8 m/s^2

Q2. A block is placed on a rough incline of angle θ . It is just about to slide. The coefficient of static friction is:

- A. $\sin\theta$
- B. $\tan\theta$
- C. $\cos\theta$
- D. $\cot\theta$

Q3. Two blocks of masses m and $2m$ are connected by a light string on a smooth horizontal surface. A force F is applied on the heavier block ($2m$). The tension in the string is:

- A. $F/3$
- B. $2F/3$
- C. $F/2$
- D. F

Q4. A block is kept inside a lift accelerating downward with acceleration g . The apparent weight of the block is:

- A. mg
- B. 0
- C. $2mg$
- D. $mg/2$

Q5. (Assertion–Reason) Assertion: A body can move in a circular path even if the net work done on it is zero. Reason: In uniform circular motion, force is perpendicular to velocity.

- A. Both correct and Reason explains Assertion
- B. Both correct but Reason does not explain
- C. Assertion correct, Reason incorrect
- D. Both incorrect

Q6. A body experiences a change in momentum linearly with time for a duration t . The force–time graph is:

- A. Constant (horizontal line)
- B. Straight line with slope

C. Exponential curve

D. Zero

Q7. Two masses m_1 and m_2 are connected over a frictionless pulley ($m_1 > m_2$). The magnitude of acceleration depends on:

A. Difference of masses only

B. Sum of masses only

C. Both sum and difference of masses

D. Only g

Q8. A block is pulled on a rough horizontal surface by a force making an angle θ above the horizontal. Friction decreases because:

A. Weight decreases

B. Normal reaction decreases

C. Area of contact decreases

D. Mass decreases

Q9. Which statement about action–reaction pairs is correct?

A. They cancel each other always

B. They act simultaneously on different bodies

C. They act on the same body

D. They have different magnitudes

Q10. A particle moves with constant speed in a circle. Which statement is correct?

A. Acceleration is zero

B. Net force is zero

C. Velocity is constant

D. Momentum changes

Q11. A particle moves such that $x = 2t^2 + 3t$ (SI units). The acceleration at $t = 2$ s is:

A. 4 m/s^2

B. 7 m/s^2

C. 2 m/s^2

D. 8 m/s^2

Q12. A projectile is fired with speed u at angle θ . The time of flight is 4 s ($g = 10 \text{ m/s}^2$). The initial vertical component $u \sin \theta$ is:

A. 10 m/s

B. 20 m/s

C. 40 m/s

D. 5 m/s

Q13. At the highest point of a projectile, which quantity is zero?

A. Speed

- B. Horizontal component of velocity
- C. Vertical component of velocity
- D. Acceleration

Q14. (Assertion–Reason) Assertion: Acceleration can be zero even when velocity is non-zero.
Reason: In uniform motion, velocity remains constant.

- A. Both correct and Reason explains Assertion
- B. Both correct but Reason does not explain
- C. Assertion correct, Reason incorrect
- D. Both incorrect

Q15. The slope of a velocity–time graph represents:

- A. Velocity
- B. Acceleration
- C. Displacement
- D. Jerk

Q16. The area under an acceleration–time graph between t_1 and t_2 gives:

- A. Change in velocity
- B. Displacement
- C. Force
- D. Momentum

Q17. A body is dropped from rest. Its speed after 2 s ($g = 10 \text{ m/s}^2$) is:

- A. 10 m/s
- B. 20 m/s
- C. 5 m/s
- D. 2 m/s

Q18. The relative velocity of A with respect to B is:

- A. $v_A - v_B$
- B. $v_A + v_B$
- C. $v_B - v_A$
- D. 0

Q19. For a projectile with time of flight 4 s ($g = 10 \text{ m/s}^2$), the maximum height reached is:

- A. 5 m
- B. 10 m
- C. 20 m
- D. 40 m

Q20. The horizontal range of a projectile (same level) is maximum at angle:

- A. 30°
- B. 45°

C. 60°

D. 90°

Answer Key

- Q1: B
Q2: B
Q3: A
Q4: B
Q5: A
Q6: A
Q7: C
Q8: B
Q9: B
Q10: D
Q11: A
Q12: B
Q13: C
Q14: A
Q15: B
Q16: A
Q17: B
Q18: A
Q19: C
Q20: B

Quick Insights (Rank Boosters)

- Q1: $f_{\text{max}} = \mu mg = 0.4 \times 5 \times 10 = 20 \text{ N}$. Doubling force $\rightarrow$ net = $40 - 20 = 20 \text{ N} \Rightarrow a = 20/5 = 4 \text{ m/s}^2$.
- Q2: Limiting equilibrium $\Rightarrow mg \sin \theta = \mu mg \cos \theta \Rightarrow \mu = \tan \theta$.
- Q3: $a = F/(3m)$; $T = m \cdot a = F/3$.
- Q4: $a = g$ downward $\Rightarrow N = m(g - g) = 0$.
- Q5: In UCM, force $\perp$ velocity $\Rightarrow$ no work; speed constant but direction changes.
- Q6: $dp/dt = \text{constant} \Rightarrow F$ constant.
- Q7: $a = (m_1 - m_2)g / (m_1 + m_2) \Rightarrow$ depends on both.
- Q8: Vertical component of pull reduces $N \Rightarrow$ friction μN decreases.
- Q9: Equal & opposite, act on different bodies, do not cancel on same body.
- Q10: Direction of velocity changes $\Rightarrow$ momentum vector changes.
- Q11: $x = 2t^2 + 3t \Rightarrow a = d^2x/dt^2 = 4 \text{ m/s}^2$.
- Q12: $T = 2u \sin \theta / g = 4 \Rightarrow u \sin \theta = 20 \text{ m/s}$.

- Q13: $v_y = 0$ at top; speed $\neq 0$ (v_x present); $a = g$ downward.
- Q14: Constant velocity $\Rightarrow$ zero acceleration; explains assertion.
- Q15: dv/dt is slope $\Rightarrow$ acceleration.
- Q16: $\int a \, dt = \Delta v$.
- Q17: $v = gt = 20 \text{ m/s}$.
- Q18: $v_{AB} = v_A - v_B$.
- Q19: $H = (u^2 \sin^2 \theta)/(2g)$; from $T=4 \Rightarrow u \sin \theta = 20 \Rightarrow H = 400/(20) = 20 \text{ m}$.
- Q20: Range max at 45° .